

Leuprorelin (Prostap®) for prostate cancer

Leuprorelin is a cancer drug. It is used to treat breast cancer and prostate cancer.

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What is leuprorelin (Prostap®)?

Leuprorelin is a hormonal therapy drug. It is also called Prostap[®]. Leuprorelin is used to treat prostate cancer.

It is best to read this information with our general information about [hormonal therapy](#) and [prostate cancer](#).

Leuprorelin can also be used to treat breast cancer. We have separate information about [leuprorelin for breast cancer](#).

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How leuprorelin works

Hormones are chemicals that our bodies make. Hormones act as messengers and help control how cells and organs work. Most prostate cancers need the hormone testosterone to grow. Testosterone is mainly made by the testicles.

Leuporelin reduces the amount of testosterone the body naturally produces. This can shrink the prostate cancer or stop it growing.

How leuporelin is given

You usually have leuporelin as an injection under skin (subcutaneously) every 3 months. Some people have it once a month as an injection into the muscle.

During your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Your nurse or doctor will talk to you about your treatment plan.

You usually have the injections at a GP surgery. A practice nurse gives them to you. If you are not able to visit the surgery, a district nurse can give you the injections at home.

You will also have regular blood tests. These check the levels of PSA protein in the blood. You may also have blood tests to check:

- your levels of blood cells
- your general health, such as how well your liver and kidneys are working.

If you are taking any medicines to thin your blood, tell your doctor, nurse or pharmacist. These medicines can make you bruise more easily.

Injection site

The skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker than the surrounding skin, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

When you have this treatment, you will have regular appointments with a doctor, nurse or pharmacist. Always tell them about any side effects you have. They can give you:

- drugs to help control most side effects
- advice about managing side effects.

Most side effects can be managed. But sometimes side effects are harder to control. It is important not to stop taking hormonal therapy without telling your doctor. If side effects cannot be managed, your doctor may suggest a different type of hormonal therapy.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Tumour flare

When you start taking this treatment your testosterone levels may go up for a very short time. This could cause any symptoms to get temporarily worse. This is sometimes called tumour flare. To prevent this, your doctor gives you another tablet to take before you start and for the first few weeks of treatment. Your doctor, nurse or pharmacist will explain more about this.

Hot flushes and sweats

Hot flushes are a common side effect of this treatment. During a flush, your neck and face may feel warm. If you have white skin, your skin may look red. If you have black or brown skin, your skin colour may deepen. Flushes can last up to 5 minutes. You might also have sweats or feel anxious or irritable during a flush.

Here are some things you can do to reduce the effects of hot flushes:

- Wear natural fabrics, such as cotton.
- Wear layers so you can remove clothes as needed.

- Use cotton sheets and layers of bedding you can remove.
- Try cooling pads or pillows to keep you cool.
- Keep rooms cool or use a fan.
- Have cold drinks and avoid caffeine and alcohol.

You might have fewer hot flushes as your body adjusts to this treatment. You could try therapies to help you [cope with hot flushes](#), such as:

- [talking therapies](#) like cognitive behaviour therapy (CBT)
- controlled breathing exercises
- yoga
- acupuncture.

If they do not improve, talk to your doctor. Certain drugs can help to improve hot flushes.

Flushes usually stop a few months after treatment ends. But some people continue to have them.

Muscle or joint pain

You may get pain in your muscles or joints. If this happens, tell your doctor or nurse. They can give you advice and painkillers or other drugs to help. Being physically active and keeping to a healthy weight may help with the pain. It can keep your joints flexible and reduce stress on them. If the pain does not get better, tell your doctor or nurse. They may talk to you about changing to a different hormonal drug.

Sexual effects

It is common to lose your [sex drive](#) and have erection difficulties with hormonal therapy. Hormonal therapy may also make your penis shorter and your testicles smaller.

Your doctor can prescribe drugs and treatments to help with erection difficulties. These will not increase your sex drive. But they might work, even with a low sex drive.

Things can improve a few months after you stop taking the drug. But it depends on how long you take hormonal therapy, and what other treatments you have had. Some people continue to have problems after treatment ends.

Feeling tired (fatigue)

[Feeling tired](#) is a common side effect. Try to pace yourself and plan your day so you have time to rest between activities.

Being physically active can help to manage tiredness and give you more energy. It also:

- helps you sleep better
- reduces stress
- improves your bone health.

If you feel sleepy, do not drive or operate machinery.

Weight gain and loss of muscle strength

You may gain weight and may lose some muscle strength with this treatment. [Eating healthily and being physically active](#) can help you keep to a healthy weight. Resistance exercises, such as lifting weights, may help to reduce loss of muscle strength. Ask your doctor or nurse for advice.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Feeling sick

Leuporelin can make you [feel sick](#) (nausea) but this is usually mild. If it does not improve, your doctor can give you anti-sickness drugs to help.

Mood changes

You may feel low or depressed, or have mood swings, during this treatment. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage low mood or [depression](#).

Difficulty sleeping

You may have difficulty sleeping while taking this treatment. We have more information about [managing sleep problems](#). Tell your doctor or nurse if it does not get better.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Breast swelling or tenderness

This treatment may cause swelling and tenderness of your breast tissue. This is called gynaecomastia. To prevent this, some people have 1 or more low-dose radiotherapy treatments to the chest before treatment starts. Another type of hormonal drug might be used to treat the breast swelling. Your doctor can give you more advice.

Swollen ankles or hands

You may notice you get swelling in your ankles or hands. This is caused by a build-up of fluid. Tell your doctor or nurse if this happens.

Effects on the liver

This treatment may affect how your liver works. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your liver is working.

Bone thinning (osteoporosis)

Taking this treatment for several months or more can increase the risk of [bone thinning](#). This is called osteoporosis. It can make you more likely to get a broken bone (fracture).

You may have bone density scans to check your bone health before and during treatment. These scans can diagnose osteoporosis or osteopenia (low bone density) so that it can be monitored. If your scan shows osteoporosis, your doctor usually prescribes:

- drugs to protect your bones called [bisphosphonates](#)
- calcium and vitamin D supplements.

Regular exercise where you support your body weight, like walking, running and dancing, can help look after your bones. This is called [weight-bearing exercise](#). Other types of strength exercises are also helpful. If you have any bone thinning or the cancer is affecting your bones, get advice from your cancer team first.

Eating a healthy diet and not smoking also helps take care of your bones.

Memory and concentration

You may notice [changes in your memory](#). You may also find it harder to concentrate.

To help you remember things, try using:

- reminder alarms or notes on your phone
- a diary, calendar or phone app
- a notebook you carry with you.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Effects on the heart

This treatment might increase certain risk factors for [heart problems](#). Tell your doctor if you have:

- had any problems with your heart
- high blood pressure
- high levels of cholesterol – a fatty substance in your blood.

They can monitor or treat these if needed. Talk to your doctor if you are worried about this.

Things you can do to take care of your heart include:

- not smoking
- maintaining a healthy weight
- being physically active.

Raised cholesterol level

This treatment can raise the level of cholesterol. This is a fatty substance in the blood. You may have a [blood test](#) to check the level of cholesterol. Your doctor may refer you to a dietitian for advice on diet, or prescribe medicines to control your cholesterol levels.

Effects on blood pressure

This treatment can cause changes in blood pressure. If you have high blood pressure, it will need to be checked regularly.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have [diabetes](#), your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Effects on the nervous system

Leuprorelin can affect the nervous system. Occasionally, leuprorelin can cause dizziness or drowsiness. If you feel like this, do not drive or operate machinery.

You may have pins and needles or feel tingling in your arms and legs. These symptoms are usually mild. If you notice any of these symptoms, tell your doctor or nurse.

Rarely, it can cause a more serious problem. Contact the hospital straight away on the number you have been given if you notice:

- severe or recurring headaches
- changes to your vision
- ringing in your ears (tinnitus)
- fits (seizures).

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

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- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Contraception

Your doctor, nurse or pharmacist will advise you not to [make someone pregnant](#) while having this treatment. The drug may harm the developing baby. It is important to use effective contraception during treatment and for a while after it finishes.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

Reviewed: 01 September 2024 | Next review: 01 September 2026

⚠ This content is currently being reviewed. New information will be coming soon.

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Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

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